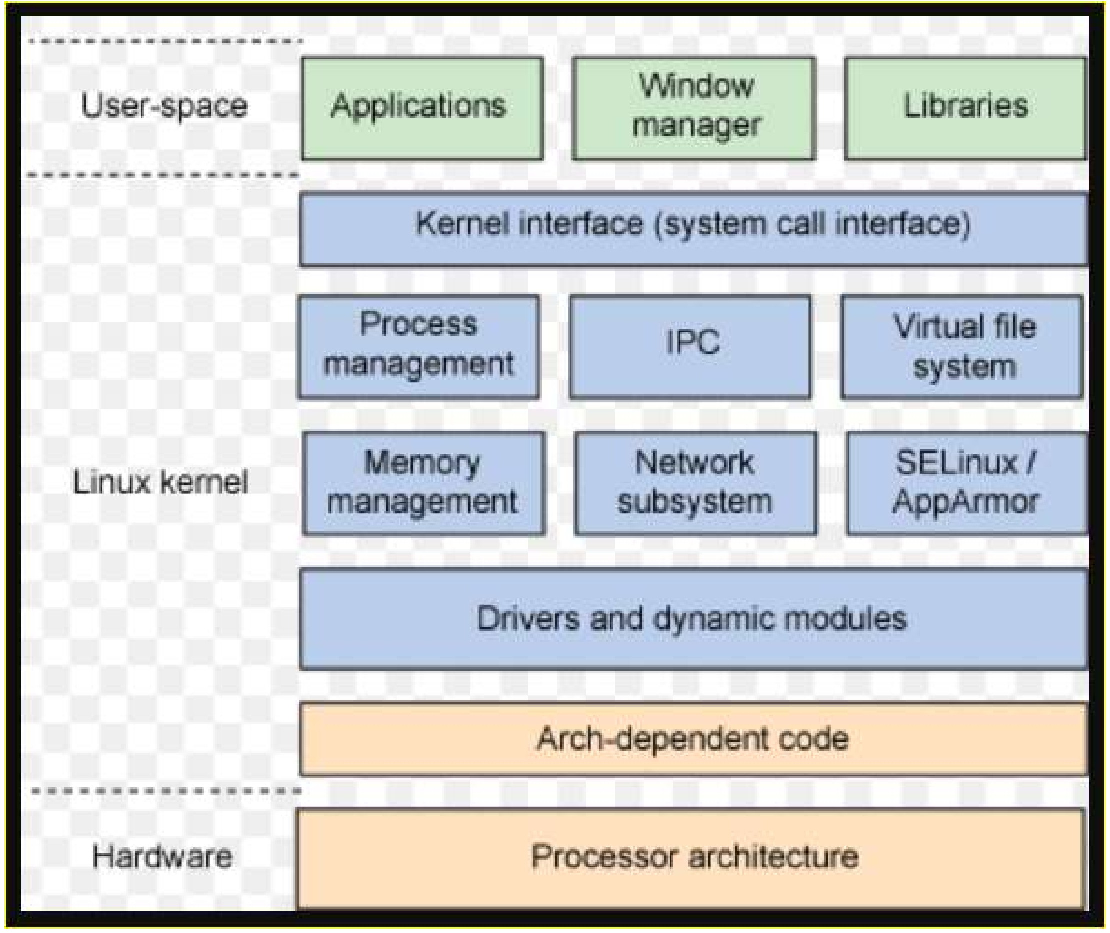
What is OS and Linux?

Linux is an open-source operating system (OS). An operating system is the software that directly manages a system's hardware and resources, like CPU, memory, and storage. The OS sits between applications and hardware and makes the connections between all your software and the physical resources that do the work.

Linux Structure



Different types of Linux Operating system available in market?
OS comes with 32bit and 64bit, below are the 3 Major Parts of OS /kernel

1. Windows
2. MacOS
3. Linux

- **Linux Flavors**

- RHEL
- CentOS
- Fedora
- Ubuntu
- SUSE Linux
- Sun Solaris

- **Diff Windows OS available in market**

<u>Desktop OS</u>

- Windows XP
- Windows Vista
- Windows 7
- Windows 10
- Windows 11

<u>Server OS</u>

- Windows Server 2000
- Windows Server 2008 and R2
- Windows Server 2012 and R2
- Widows Server 2016
- Windows Server 2019

Diff between Windows OS and Linux OS?

- Usage
- File Systems
- Security
- Compatibility
- Ease of Use
- License
- Reliability

Most of the common servers we can configure in Linux

- Samba
- FTP
- DNS
- DHCP
- HTTP/HTTPS
- Tomcat
- SVN - versioning
- Ops VPN
- Nagios
- Open LDAP
- Postfix
- PAM
- HAProxy
- MySQL
- PostgreSQL
- Mongo
- Cassnadra

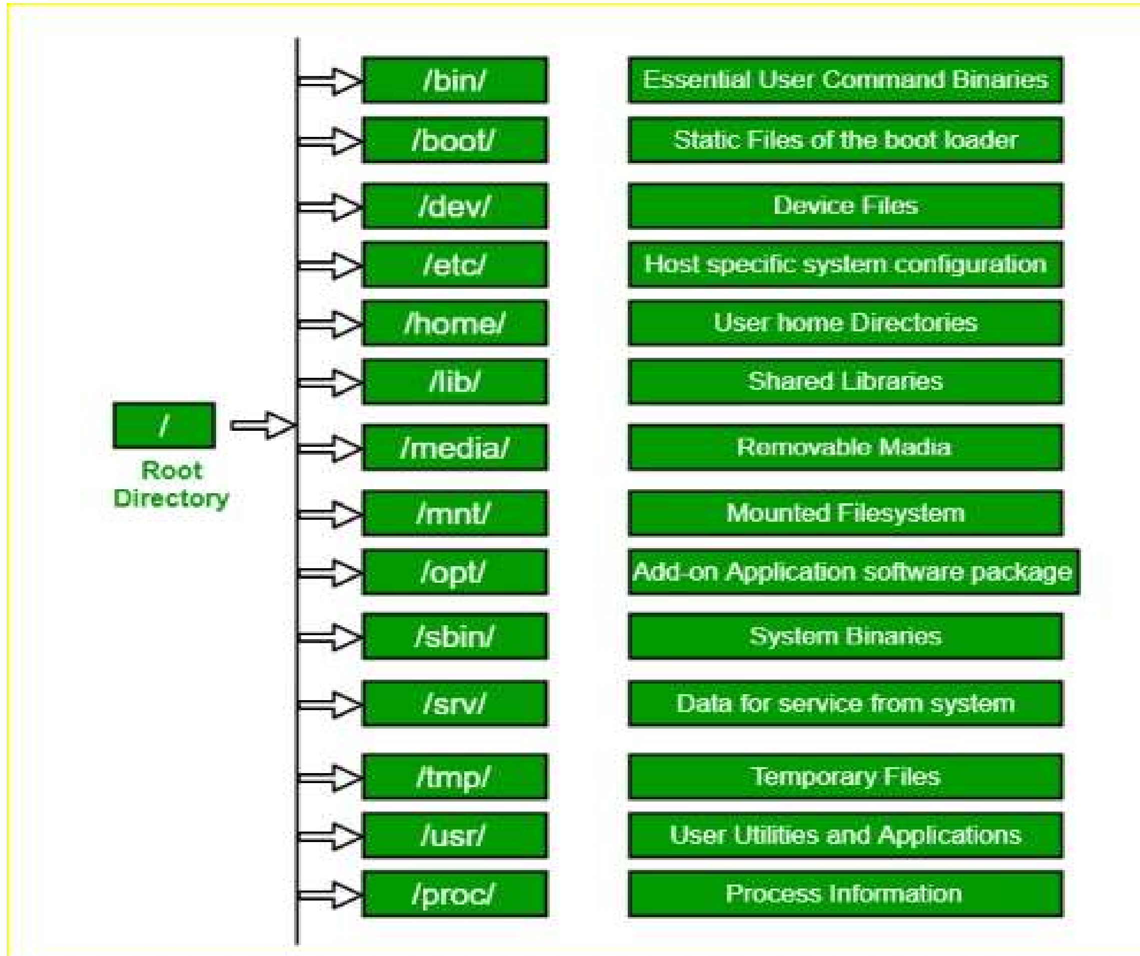
Standard Ports in Linux?

There are 65,535 possible port numbers, although not all are in common use.
Some of the most used ports, along with their associated networking protocol

- **Ports 20 and 21:** File Transfer Protocol (FTP). FTP is for transferring files between a client and a server.
- **Port 22**: Secure Shell (SSH). SSH is one of many tunnelling protocols that create secure network connections.
- **Port 25:** Simple Mail Transfer Protocol (SMTP). SMTP is used for email.
- **Port 53:** Domain Name System (DNS). DNS is an essential process for the modern Internet; it matches human-readable domain names to machine-readable IP addresses, enabling users to load websites and applications without memorizing a long list of IP addresses.
- **Port 80:** Hypertext Transfer Protocol (HTTP). HTTP is the protocol that makes the World Wide Web possible.
- **Port 123:** Network Time Protocol (NTP). NTP allows computer clocks to sync with each other, a process that is essential for encryption.
- **Port 179:** Border Gateway Protocol (BGP). BGP is essential for establishing efficient routes between the large networks that make up the Internet (these large networks are called autonomous systems). Autonomous systems use BGP to broadcast which IP addresses they control.
- **Port 443:** HTTP Secure (HTTPS). HTTPS is the secure and encrypted version of HTTP. All HTTPS web traffic goes to port 443. Network services that use HTTPS for encryption, such as DNS over HTTPS, also connect at this port.
- **Port 3389**: Remote Desktop Protocol (RDP). RDP enables users to remotely connect to their desktop computers from another device.

The Filesystem Hierarchy Standard

It defines the directory structure and directory contents in Unix-like operating systems. It is maintained by the Linux Foundation.



1. **/ (Root):** Primary hierarchy root and root directory of the entire file system hierarchy.

2. **/bin** : Essential command binaries that need to be available in single-user mode; for all users, e.g., cat, ls, cp.

3. /boot : Boot loader files, e.g., kernels, initrd.

4. /dev : Essential device files, e.g., /dev/null.

5. /etc : Host-specific system-wide configuration files.

6. /home : Users' home directories, containing saved files, personal settings, etc.

7. /lib : Libraries essential for the binaries in /bin/ and /sbin/.

8. /media : Mount points for removable media such as CD-ROMs (appeared in FHS-2.3).

9. /mnt : Temporarily mounted filesystems.

10. /opt : Optional application software packages.

11. /sbin : Essential system binaries, e.g., fsck, init, route.

12. /srv : Site-specific data served by this system, such as data and scripts for web servers, data offered by FTP servers, and repositories for version control systems.

13. /tmp : Temporary files. Often not preserved between system reboots, and may be severely size restricted.

14. /usr : Secondary hierarchy for read-only user data; contains the majority of (multi-)user utilities and applications.

15. /proc : Virtual filesystem providing process and kernel information as files. In Linux, corresponds to a procfs mount. Generally automatically generated and populated by the system, on the fly.

What is repository in Linux

- Yum
- APT

- Installation of CentOS Link

https://www.youtube.com/watch?v=NJYzYClIwxw

- Configuration of Virtual Box link

https://www.youtube.com/watch?v=ovgHSr0OZAk